



SOLUTIONS FOR A CLEANER TOMORROW

CLEAN AIR
SAFER INDUSTRY
BRIGHTER TOMORROW

VERTICAL BIO-SCRUBBER LNS-SC-BIO

Industrial Gas Cleaning &
Biological Odour Control System



**Reliable
Operation**
Engineered for
continuous performance



**Corrosion-Resistant
Design**
Built for harsh
industrial environments



**Sustainable
Odour Control**
Harnessing natural
biological treatment

PEOPLE
PROCESS
A CLEANER
PLANET

Chemical Dosing System
For Biological Regeneration
• Nutrient Dosing
• pH Control (Optional)



AIR | PEOPLE | PROGRESS

INDUSTRIAL BIO-SCRUBBER SYSTEM

1. PROCESS OVERVIEW:

The Industrial Bio scrubber System is an advanced, two-stage environmental pollution control process engineered for the continuous extraction and biological degradation of high-volume water-soluble volatile organic compounds (VOCs) and hazardous odorous process emissions. The system seamlessly couples a high-efficiency vertical counter-current absorption tower (scrubber) with an integrated biological oxidation reactor (sump/regeneration tank). Contaminated process gas is introduced at the column base and driven upward through a high-surface-area packing media, where target pollutants are aggressively mass-transferred into a falling chemically and physically balanced aqueous phase. The loaded scrubbing liquid drains directly into the bioreactor, where a dense population of specialized, acclimated aerobic microorganisms continuously breaks down and metabolizes the toxic pollutants into harmless, environmentally benign end products, primarily Carbon Dioxide (CO₂), Water (H₂O), and basic mineral salts.

2. INDUSTRIAL APPLICATIONS:

The system is optimized for continuous operation across highly corrosive and biologically intensive industrial profiles:

• Municipal Wastewater Systems (WWTP):

Abatement of aggressive headspace odors from headworks, primary clarifiers, primary thickeners, and mechanical sludge dewatering galleries. Targets heavy concentrations of Hydrogen Sulfide (H₂S), methyl mercaptans, dimethyl sulfides, and complex organic amines.

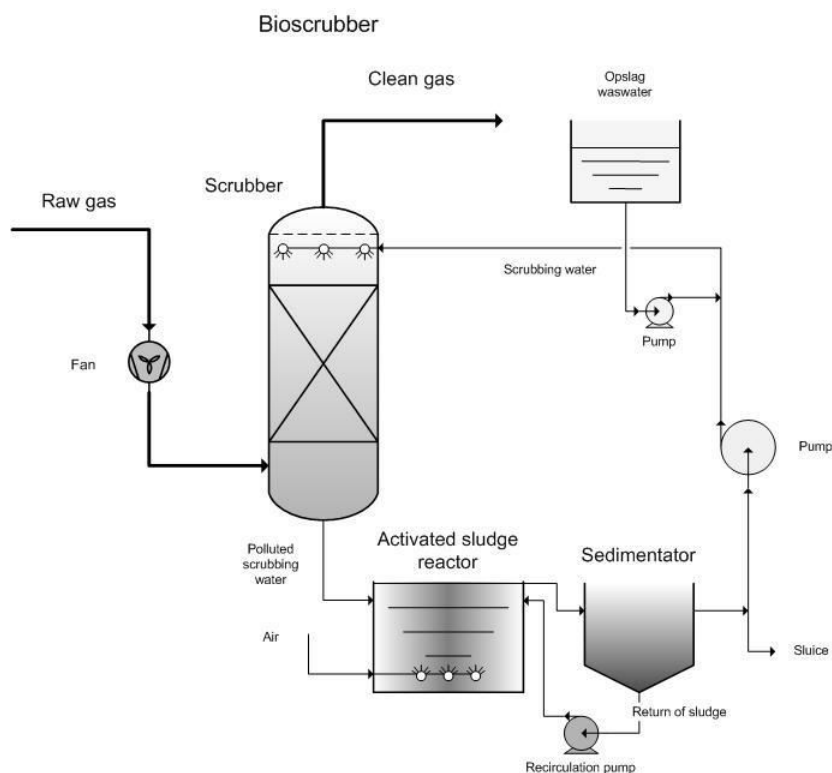
• Chemical & Pharmaceutical Process Lines:

High-yield containment of water-soluble organic process emissions, solvent storage tank vents, and reaction vessel exhaust streams. Specifically targets concentrated mass streams of Methanol, Ethanol, Acetone, Formaldehyde, and low-molecular-weight esters.

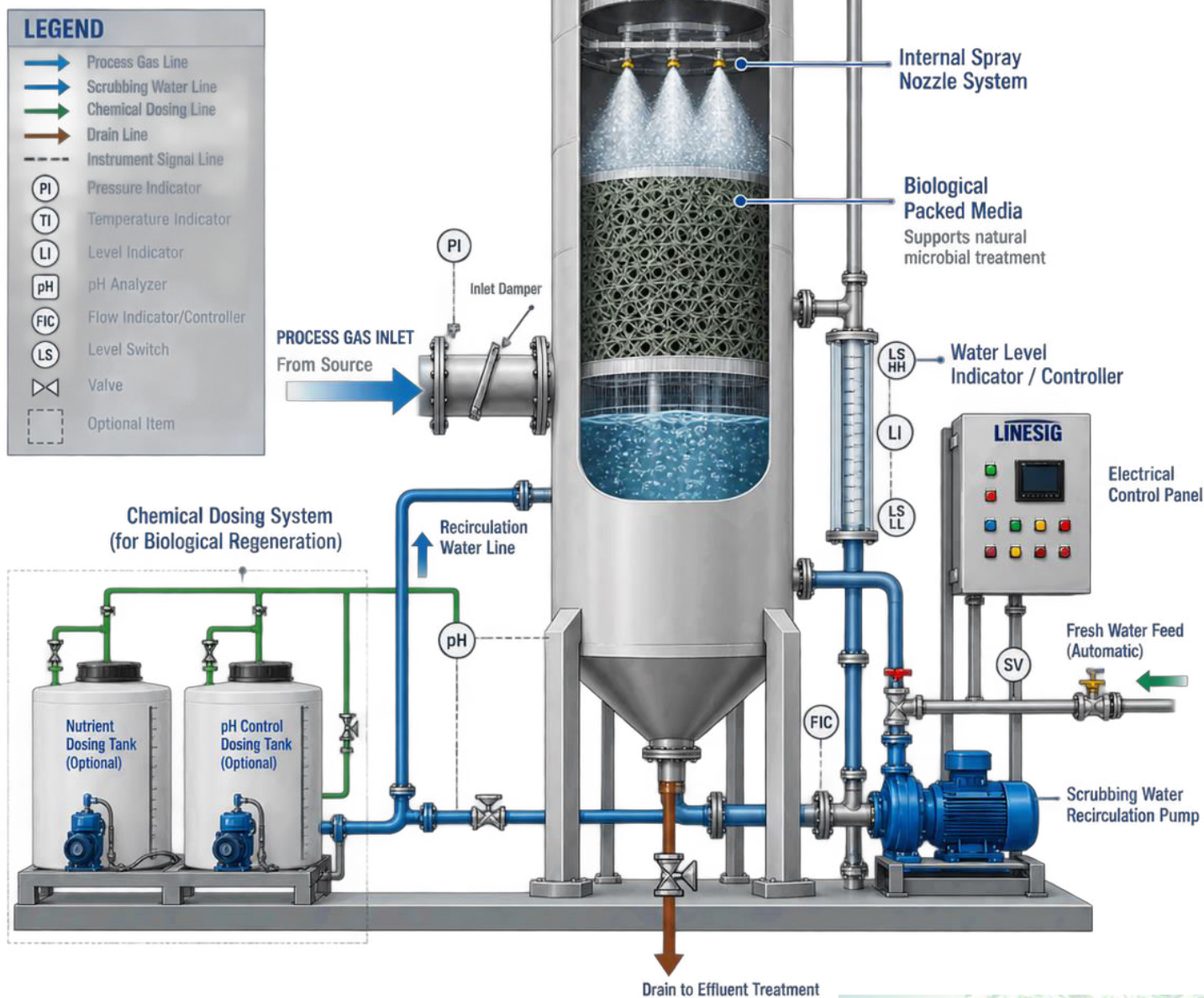
• Food Processing, Rendering &

Fermentation Lines: Treatment of intense off-gases generated from animal rendering cookers, flash dryers, commercial food preparation facilities, and industrial fermentation vents. Targets volatile fatty acids (VFAs), ammonia, trimethylamines, and complex organic odors.

Pulp & Paper Mills: Neutralization of continuous low-concentration / high-volume Non-Condensable Gas (NCG) streams containing reduced organic sulfur fractions, mercaptan matrices, and localized process chemical fumes.



SCHEMATIC DIAGRAM



3. Technical & Performance Specifications

Parameter	Nominal Value / Design Criterion	Engineering Comments
Nominal Volumetric Gas Flow Rate	10,000 m³/h (5,885 CFM)	Sized for typical industrial exhaust manifold constraints
Empty Bed Residence Time (EBRT)	2.0 – 6.0 seconds	Optimized for rapid mass transfer across structural packing
Liquid-to-Gas (L/G) Ratio	3.5 – 4.5 L/m³	Guarantees continuous wetting and eliminates dry channeling
Max Packing Differential Pressure Drop	< 250 Pa / m of media bed	Minimizes gas blower static loading and power draw
Bioreactor Hydraulic Retention Time (HRT)	4.0 – 8.0 hours	Provides sufficient contact time for complete bio-oxidation
Mixed Liquor Suspended Solids (MLSS)	1,000 – 3,000 mg/L	Maintains ideal active biological density without clogging
Target Contaminant Removal Efficiency	Up to 99.9% (H ₂ S / Soluble VOCs)	Maintains strict compliance with stringent local emission caps

4. ABSORBER COLUMN MECHANICS:

Shell Construction:

Prefabricated from industrial premium UV-stabilized Fiber Reinforced Plastic (FRP) utilizing high-grade Vinyl Ester Resin liners to provide complete chemical immunity against acidic biogenic byproducts.

Packing Elements:

High-performance structured or randomized Polypropylene (PP) packing block matrices boasting an active mass-transfer specific surface area >150 m²/m³.

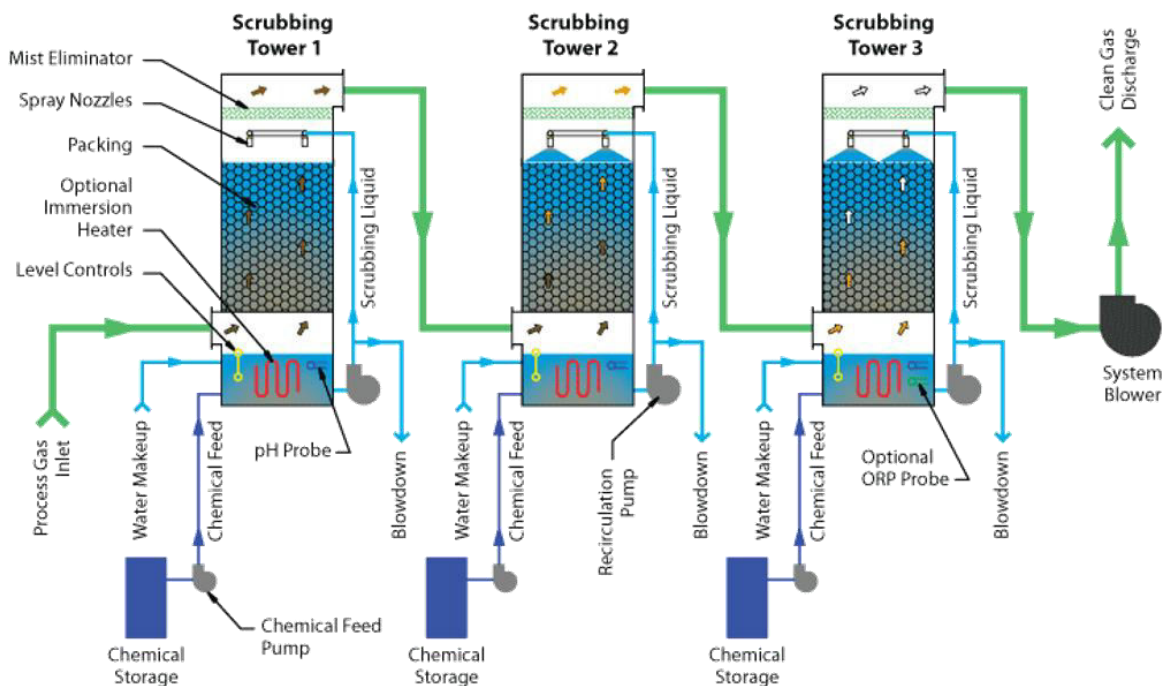
Liquid Distribution Network:

Fully integrated PVC/PP internal header pipes fitted with non-clogging, heavy-duty full-cone spray nozzles engineered for complete, homogenous overlapping cross-sectional hydraulic coverage.

Demister Assembly:

Positioned above the distribution grid; high-efficiency PP Chevron-vane profile mist eliminator designed to trap 99% of entrained water droplets >10 microns before stack discharge.

Component	Mechanical Specification & Design Features
Shell Material (MOC)	UV-stabilized Fiber Reinforced Plastic (FRP) with premium Vinyl Ester resin liner
Packing Medium Type	Random or structured high-efficiency Polypropylene (PP) rings; specific surface area >150 m²/m³
Liquid Distributor	Non-clogging, full-cone wide angle pigtail spray nozzles mounted on a removable PP header piping grid
Mist Eliminator	High-efficiency triple-track Chevron vane demister profile, PP material, captures droplets >10 microns



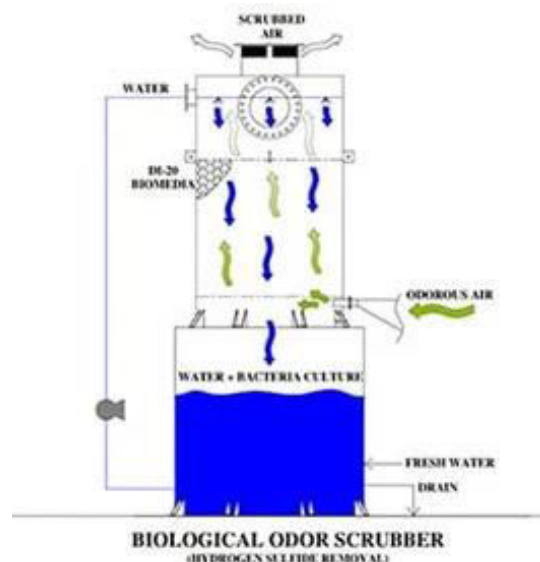
5. BIOLOGICAL REACTOR & REGENERATION TANK

Submerged Diffuser Matrix:

Heavy-duty floor-anchored EPDM fine-bubble disc or tube diffuser grid engineered to deliver highly uniform, continuous micro-bubbles. Maximizes Standard Oxygen Transfer Efficiency (SOTE) and generates sufficient upward velocity to keep biological sludge in continuous suspension, preventing anaerobic localized zones.

Aeration Air Blower Skid:

Independent, dedicated positive-displacement tri-lobe or rotary claw blower skid, driven by a VFD-modulated premium efficiency motor. Blower speed is automated via closed-loop feedback from submerged online Dissolved Oxygen (DO) galvanic sensors to dynamically maintain a fixed operational target of **2.0 – 4.0 mg/L DO**.



Parameter / Component	Biological Reactor Specification
Hydraulic Retention Time	4.0 – 8.0 hours based on biological load and pollutant kinetics
Biomass Concentration	1,000 – 3,000 mg/L Mixed Liquor Suspended Solids (MLSS)
Dissolved Oxygen Loop	2.0 – 4.0 mg/L constant residual maintained via variable frequency drive (VFD) blowers
Aeration Grid Infrastructure	Floor-mounted fine-bubble EPDM disc diffusers arranged for complete mix hydraulics

6. PROCESS GAS BLOWER SUBSYSTEM

Configuration & Material:

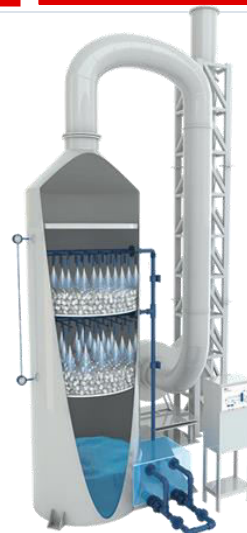
Heavy-duty, high-efficiency centrifugal direct-drive process blower with an aerodynamic molded, anti-corrosive PP or solid FRP casing and precision-balanced dynamic impeller assembly.

Operational Thresholds:

Rated for a nominal volumetric capacity of 10,000 m³/h (5,885 CFM) while generating a total static pressure of 2,500 – 3,500 Pa (10 – 14 in. WG) to easily overcome packing resistance, mist eliminators, and external duct runs.

Electrical & Area Safety:

Equipped with a VFD-compatible TEFC motor rated for ATEX Zone 2 / Class I Div 2 spark-proof hazardous locations.



Equipment Subsystem	Detailed Engineering Parameters
Process Gas Blower Type	Heavy-duty Industrial Centrifugal Blower (Direct or Belt Driven)
Gas Blower Volumetric Flow	10,000 m³/h continuous rating
Gas Blower Static Pressure	2,500 – 3,500 Pa (10 – 14 in. WG) total system static pressure resistance
Gas Blower Construction	Anti-corrosive solid PP or FRP casing with carbon-loaded ATEX spark-proof impeller
Scrubbing Water Pump Type	Horizontal, single-stage end-suction centrifugal pump assembly
Scrubbing Water Pump Flow	35 – 45 m³/h (154 – 198 GPM) continuous delivery profile
Total Dynamic Head (TDH)	2.5 – 3.5 bar (25 – 35 meters water column)
Pump Material & Seals	Solid molded PP or PVDF body; Silicon Carbide vs. Silicon Carbide slurry mechanical seals
Redundancy Matrix	1 Duty + 1 Standby (100% redundant configuration with auto-changeover logic)

7. SCRUBBING WATER RECIRCULATION PUMP SUBSYSTEM

Design Architecture:

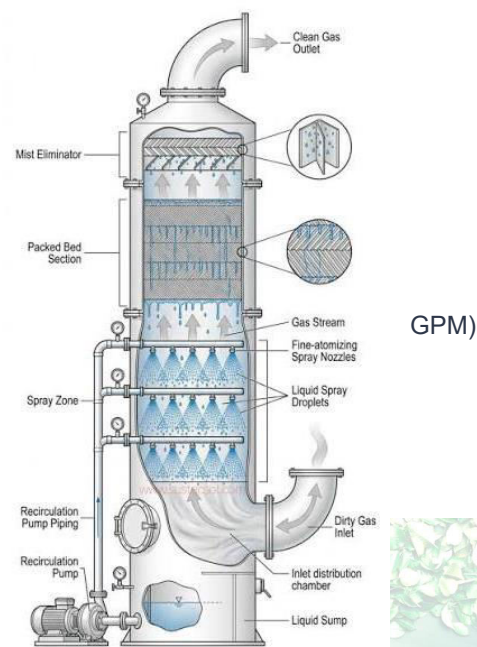
Horizontal end-suction centrifugal pump assemblies arranged in an uncompromised **1 Duty + 1 Standby (100% Redundancy)** mechanical configuration to guarantee uninterrupted 24/7 continuous operation.

Performance Metrics:

Engineered to deliver a constant volumetric liquid flow of 35 – 45 m³/h (154 – 198 at a Total Dynamic Head (TDH) of 2.5 – 3.5 bar (25 – 35 m w.c.) to properly energize upper column atomization nozzles.

Metallurgy & Mechanical Seals:

Solid injection-molded thick-walled PP, PVDF, or PFA lined volutes and impellers. Outfitted with robust, double-acting abrasive-resistant Silicon Carbide vs. Silicon Carbide (SiC/SiC) fluid-flushed mechanical seals.

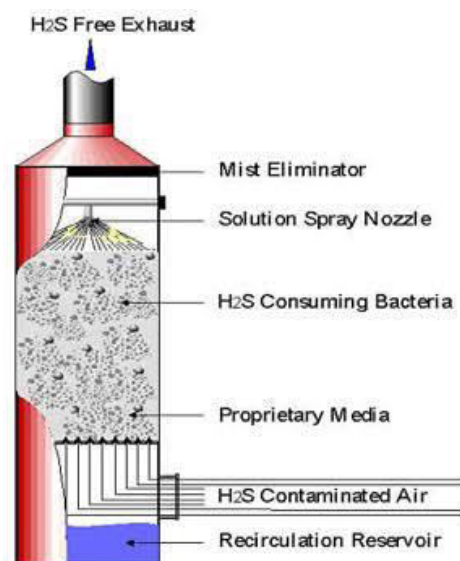


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8. CHEMICAL INJECTION SYSTEM & REAGENT MATRIX

Pump Specifications: High-precision, heavy-duty positive displacement electromagnetic or motor-driven diaphragm metering pumps. Liquid ends are machined from solid PVDF with spring-loaded ceramic check valves and pure, double PTFE-faced diaphragms to eliminate structural corrosion paths.

Integration & Dosing Logic: Pumps receive a direct 4-20mA proportional-pacing signal from the master PLC panel. Chemical delivery rates are modulated in real-time based on continuous inline inputs from upstream process instrumentation.



System Element	Engineering & Process Chemistry Specifications
Injection Pump Type	Positive displacement, high-precision electromagnetic or motor-driven diaphragm metering pumps
Pump Material of Construction	PVDF liquid end, PTFE-faced diaphragm, ceramic valves for universal chemical resistance
Control Configuration	4-20mA proportional pacing modulated by online pH, ORP, or EC transmitters via central PLC
Redundancy Scheme	Dedicated independent pumps per dosing channel with manual block-and-bleed bypass manifolds
Reagent 1: pH Correction (Acid)	Sulfuric Acid (H ₂ SO ₄ , 93-98%): Dosed to counteract alkaline drift caused by heavy ammonia (NH ₃) absorption.
Reagent 2: pH Correction (Alkali)	Sodium Hydroxide (NaOH, 25-50%): Dosed to neutralize acidic oxidation bioproducts generated from Hydrogen Sulfide (H ₂ S) degradation.
Reagent 3: Macronutrients	Aqueous Nitrogen & Phosphorus Blend: Commercial formulations (e.g., Urea/Phosphoric acid mixes) to maintain optimal microbial C:N:P metabolic balance.
Reagent 4: Supplemental Biocides	Sodium Hypochlorite (NaOCl, 12.5%) / Hydrogen Peroxide: Intermittent shock dosing capability reserved for biological fouling control or packing sterilization during turnarounds.

9. INTEGRATED PROCESS CONTROLS & INSTRUMENTATION

The system features a centralized, fully automated control panel built around a high-performance programmable logic controller (**Siemens S7-1500 or Allen-Bradley ControlLogix** architecture) housed in an IP65/NEMA 4X stainless steel enclosure.

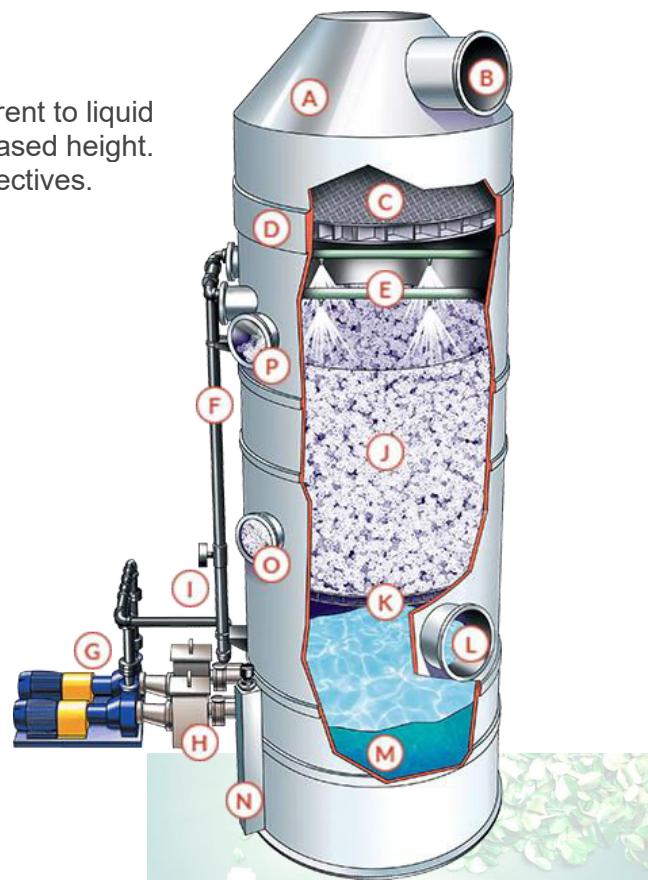
- **pH Control Loop:** Dual redundant online glass pH electrodes continuously sample the bioreactor. If pH drops below 6.8 (acidic generation), the NaOH dosing pump initiates. If pH exceeds 7.8 (alkaline buildup), the H2SO4 pump is engaged.
- **Automated Blowdown System:** An online Electrical Conductivity (EC) sensor monitors the accumulation of dissolved salts and byproducts in the sump liquid. Upon breaching a threshold (e.g., 5,000 $\mu\text{S}/\text{cm}$), a motorized blowdown ball valve opens to purge the salt-laden liquor, while an ultrasonic level sensor initiates fresh make-up water valves to maintain a stable tank volume.
- **Fouling & Safety Interlocks:** Differential pressure transmitters continuously map the pressure drop across the packed bed. An alarm is triggered if the delta-P crosses 250 Pa/m, alerting operators to bio-fouling or chemical scaling. Differential pressures are also linked to safety shutoffs that instantly interlock chemical lines if recirculation water flow drops below nominal ratings.

Control Loop	Automation & Instrument Strategy
Central Control Unit	Siemens S7-1200/1500 or Allen-Bradley Micro850/CompactLogix PLC architecture with Profinet/Modbus IP
pH Control Loop	Dual-redundant immersion pH probes activating proportional chemical dosing pumps to keep range within 6.8 – 7.8
Blowdown / Purge Loop	Automated blowdown valve modulated by liquor electrical conductivity (EC) or turbidity to control accumulated salts and sludge

10. SYSTEM CONFIGURATION

Packed bed scrubbers' vertical format allows gas to flow counter-current to liquid flow packed scrubbers have a smaller footprint in exchange for increased height. They're generally less costly from both capital and operational perspectives.

A	Scrubber body
B	Outlet from the scrubber
C	Mist eliminator
D	Mist eliminator access door
E	Recirculation liquid distribution nozzle system
F	Recirculation piping system
G	Recirculation pumps
H	(Optional) recirculation system basket strainer
I	Liquid flow meter
J	Packing media: Tri-Packs
K	Packing support grid
L	Scrubber inlet
M	Recirculation tank
N	Recirculation sump level control system
O	Packing removal portal
P	Packing fill door





CLEANER INDUSTRY. HEALTHIER AIR. GREENER TOMORROW.

Advanced scrubber solutions for responsible industrial emission and waste-air treatment.